

TEST

DATA HANDLING
AND VISUALIZATION

- 1) Explain the concept of mapping data onto aesthetics in data visualization. Discuss how different types of data are represented using visual aesthetics position, size, shape and colour. Illustrate your answer with suitable examples.

Mapping data onto aesthetics is the process of assigning data values to the form of visualization. Which is to make the user understand the data easily. Complex data are made easily readable using data visualization. Visual aesthetics such as position, size, shape and colour is used to differentiate the types of data.

VISUAL AESTHETICS:

* Position:

It is used to represent whether the data lies in the x-axis or y-axis. Data in numerical forms are mostly represented using this position.

* Size:

It is used to represent the quantity of the data. The more the quantity the larger the size is.

* Shape:

It is used to represent the categories of the data. It is easy to identify and understand different categories.

* Colour:

It is used to represent the difference or the intensity of the data.

Categorical:

Datas that are grouped into different types without any order.

Visual Aesthetics: Position, size, colour.

Example: Different departments under sales.

Ordinal:

Datas that represent the positive and negatives of the values.

Visual aesthetics: Colour intensity, shape

Example: Customer service review (Good, okay, Bad)

Quantitative:

Datas in numerical values which are measurable.

Visual Aesthetics: position, size, colour.

Example: Temperature (Celsius or Fahrenheit).

2) Role of Scales in Data Visualization.

It is used to convert data into visualization form.

Role of Scale:

- It converts data to visualization form.
- * It gives accurate values of the data.
- It gives effective readability.
- * It helps to compare data values for interpretation.

Types of Scales:

- * Linear Scale
- * Logarithmic Scale
- * Ordinary Scale
- * Colour Scale.

Linear Scale:

It represents data in a sequence form.

Example: Company Sales data.

Logarithmic Scale:

This scale is used for larger data types.

Example: Temperature, population.

Ordinary Scale:

Data is in a order or arranged

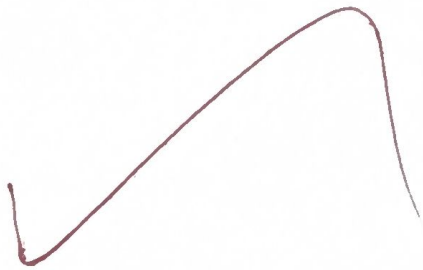
form.

Example: Low, medium, high.

Colour Scale:

Data is used to differentiate data into categories.

Example: Student detail records.



5) Describe the role of colour in data visualization. Differentiate between using colour to distinguish categories, represent data values and highlight important information. Discuss the principles of choosing effective colour palettes, including contrast, accessibility and visual harmony, and explain how poor colour choices can affect data interpretation.

Colour plays an important role in data visualization. It is used to differentiate types of data. It differentiates or groups data values into different categories or groups.

Colour is used to categorize data, for example: School Records:

Student marks, Student IDs, Student names, Student contacts.

Colour is useful to make the user to understand and differentiate and categorize data easily.

Colour represents data values, for example: sales department, marketing department. Colour is also used to represent the data values.

Colour palettes:

Choosing the effective colour palettes is important, that represents the intensity of the data.

Example: Customer Review

Bad → Red

Okay → Yellow

Good → Green

Choosing effective colour palettes represents the data and data values correctly.

Poor Colour Choices:

Poor colour choices can affect the data visualization. It is difficult to understand the accurately data values may not reach the user effectively. The data may be inappropriate if the colours are poorly chosen.

It gives a poor data visualization or graphs, which are ineffective.